

GIHEKE TSS
END OF TERM I EXAMINATIONS
SECTOR: ICT & MULTIMEDIA
TRADE: L3 NIT
ACADEMIC YEAR: 2024/2025
MODULE TITLE: NETWORK FUNDAMENTAL
MAX: 100 Marks
DATE: 10/12/2024
DURATION: 9h00-12h00

INSTRUCTIONS TO CANDIDATES:

This Exam paper is composed of Three ,Sections (A, B,C).
Follow the instructions given below, and answer the indicated
Trace the line after responding each question

Section A:

Section B:

Section C:

SECTION A: Attempt All questions(55marks)

01	Define computer network	2marks										
02	Outline elements of computer network	4marks										
03	What is protocol	3marks										
04	What do you understand by the following protocols I. DHCP II. TCP III. Apple Talk IV. OSPF	4marks										
05	Outline the types of transmission media and two(2) examples for each	4marks										
06	Describe the types of network standard	4marks										
07	Based on the class of network devices belong in the table below, give at least two (2) examples for each class <table><tr><td>Classes of network devices</td><td>Examples</td></tr><tr><td>Access devices</td><td></td></tr><tr><td>Interconnection Devices</td><td></td></tr><tr><td>End Devices</td><td></td></tr><tr><td>Security devices</td><td></td></tr></table>	Classes of network devices	Examples	Access devices		Interconnection Devices		End Devices		Security devices		4Marks
Classes of network devices	Examples											
Access devices												
Interconnection Devices												
End Devices												
Security devices												
08	List three (3) major types of signals transmitted through logical media	3 marks										
09	Show the use of baseband and broadband	4marks										
10	List ten(10) application of network in real life	5marks										
11	Choose the correct answer a) Which protocol is used to Automatically provide ip address to the computer? I. DNS	4marks										

	II. DHCP III. ARP IV. IGMP b) Which protocol is used to provide secure connection across the internet I. ARP II. HTTPS III. NTP IV. POP3	
12	Write in full form A. TCP/IP B. DNS C. HTTP D. FTP	2mmarks
13	What is network topology	2Marks
14	Answer by True when the statement is correct and False when statement is not correct i. Local area network is designed to operate over a large geographical area such as a country or continents ii. Metropolitan area network use similar technologies as Lan but span over a large geographical such as town or entire city iii. One of the advantages of wide area network is that it is costlier to implements iv. Resource sharing is not possible advantages of a local area network(Lan)	2marks
15	Which of the following is a function of a router? A) Connects devices within the same network B) Filters network traffic C) Connects different networks and routes data between them D) Amplifies signals	2marks
16	As network designer you are hired for implementing mesh topology which composed by 50 nodes so how can respond the following questions a)number of all port in the network?/ b)number of port for each node? c)number of cables for entire network?	4marks
17	which layer of OSI model which handle error detection in communication?	2marks

Section B: Attempt any three (3) questions (30 marks)

18	Compare OSI model layer from TCP/IP model layers	(10 marks)
19	As a trainee who learnt this module of network describe classification on network based on geographical area and based on components roles	(10 marks)
20	Compare baseband and broadband transmission Technologies	(10 marks)
21	By using table differentiate Radio waves, infrared ,Microwaves ,wi-fi as wireless transmission techniques	(10 marks)
22	Draw any four(5) network topology you know and provide two advantages and disadvantages for each?	(10 marks)

Section C: Attempt only one (1) question (15 marks)

23	Suppose you are requested to build a network topology that has a central point. Each node attached to central point has a dedicated connection A. Draw and explain that topology B. Give three (3) advantages and three (3) disadvantages of that topology	(15marks)
24	As technician hired by bright solution company located at Nyanza to fix the problem of having insufficient network cable and they decide to make new cables, what are the steps should be followed during the process of making new cable	(15marks)

END OF ASSESSMENT